



AMF STUDIO

PROJECT SOURCE DOCUMENTATION

Prepared By:	KIRUBAKARAN V
Organization:	CODETHRIVE INFOTECH / AMF STUDIO
Technology Stack:	Full-Stack Web (HTML5, Custom CSS3, JS, Vite, Node.js, Express.js, PostgreSQL / Neon)
Date:	08/06/2026 – 02/07/2026
Live URL:	https://amfstudio.in/

1. PROJECT OVERVIEW

AMF STUDIO is a modern, high-performance web-based creative production platform designed for photography, cinematic videography, audio recording, and event production services. The platform provides patients and clients with an intuitive interface to explore creative services, view filterable portfolio galleries, book shoot time slots in real-time, and submit project inquiries. An Admin Management Dashboard enables studio operators to track, filter, and manage all bookings securely stored in a cloud PostgreSQL database.

2. PROJECT OBJECTIVES

- **Seamless Online Booking:** Provide real-time date and time-slot selection for photography and videography sessions.
- **Comprehensive Portfolio Showcase:** Display categorized high-definition photos and videos for weddings, events, pre-weddings, and corporate shoots.
- **Transparent Service Offerings:** Present detailed shoot packages, custom add-ons, and pricing structures clearly.
- **Automated Cloud Data Persistence:** Securely store client bookings and inquiry entries in a scalable PostgreSQL cloud database (Neon).
- **Modern & Fast UI/UX:** Deliver a mobile-responsive interface featuring smooth glassmorphism design, keyframe animations, and dark mode.
- **Efficient Admin Control:** Provide studio management with a secure admin dashboard to search, filter, paginate, and manage client bookings.

3. PROJECT SCOPE

The project encompasses client-facing multi-page web applications, real-time shoot scheduling modules, filterable portfolio media galleries, inquiry request portals, Node.js REST API endpoints, cloud PostgreSQL database management, and a secure Admin Dashboard for session scheduling and client communications.

4. REQUIREMENTS

4.1. FUNCTIONAL REQUIREMENTS

- **Home Page:** Hero showcase with dynamic animated text rotator, service highlights, process workflow, client reviews, and quick booking options.
- **About Us:** Studio background, vision & mission, founder/team profiles, camera/lighting equipment specs, and creative philosophy.
- **Services:** Detailed overview of packages (Wedding, Pre-Wedding, Maternity, Puberty, Corporate, Model Shoots, Product Ad Shoots).

- **Portfolio:** Interactive photo/video gallery with instant category filtering (Weddings, Pre-Weddings, Events, Commercial).
- **Booking Module:** Real-time appointment scheduling with date picker, dynamic slot fetcher, service category selection, and double-booking prevention.
- **Contact Page:** Studio contact details, inquiry form, interactive Google Maps location embed, emergency contact CTAs, and social links.
- **Admin Panel:** Secure dashboard for viewing, searching, filtering (by category/name), paginating, and deleting customer booking records.

4.2. NON-FUNCTIONAL REQUIREMENTS

- **Responsive Design:** Fully adaptive layout optimized across mobile phones, tablets, laptops, and desktop screens.
- **Fast Performance:** Sub-second page rendering powered by Vite build optimization and WebP image compression.
- **Secure Data Handling:** Parameterized SQL queries protecting against SQL injection vulnerabilities and form input validation.
- **Cross-Browser Compatibility:** Seamless performance on Chrome, Edge, Safari, Firefox, and mobile web browsers.
- **Reliable Cloud Backend:** Express REST API connected with Neon Serverless PostgreSQL featuring SSL encryption and error recovery.

5. TECHNOLOGY STACK & MODULES

Component	Technology / Specification
Frontend Core	HTML5, JavaScript (ES6+ ESNext)
Build Tool	Vite (v5)
Styling & Design System	Custom Vanilla CSS3 (Design Tokens, Glassmorphism, CSS Grid, Animations)
Icons	Font Awesome 6 (Free CDN)
Backend Framework	Node.js, Express.js (v5)
Database	PostgreSQL (Neon Cloud Serverless DB)
Database Client	`pg` (Node-PostgreSQL Connection Pool)
Architecture	Full-Stack Web App (Vite Static Frontend + Express REST API + Cloud PostgreSQL)
API Specification	RESTful JSON APIs (`/api/bookings`)

6. SYSTEM MODULES

- **Home Module:** Intro showcase, animated text rotator, service highlights, process workflow, client testimonials.
- **About Module:** Studio heritage, equipment list, creative vision, founder credentials.
- **Services Module:** Categorized photography & videography offerings, customizable shoot add-ons.
- **Portfolio Module:** Filterable media gallery backed by high-resolution optimized imagery.
- **Booking & Slot Picker Module:** Date picker, dynamic time slot fetcher, category dropdown, POST ``/api/bookings`` submission.
- **Contact & Inquiries Module:** General client query form, location map embed, direct phone helpline.
- **Admin Dashboard Module:** Tabbed table interface for managing bookings, live search, category filter, pagination, DELETE ``/api/bookings/:id``.

7. USER ROLES

CLIENTS / VISITORS

- Browse photography and videography services, portfolio galleries, and studio equipment details.
- Book in-person shoot sessions with date and time-slot selection.
- Submit project inquiry forms with custom shoot requirements.
- View studio location, contact details, and client testimonials.

STUDIO MANAGEMENT / ADMIN

- Access secure Admin Portal via protected URL route.
- View, search, and filter all customer session bookings in real-time.
- Filter bookings by shoot category, customer name, or email address.
- Paginate through large sets of booking entries efficiently.
- Delete completed, cancelled, or duplicate booking entries.

8. SECURITY

- **Parameterized Database Queries:** All PostgreSQL interactions use parameterized ``$1, $2, ...`` queries to protect against SQL Injection.
- **Environment Variable Protection:** Sensitive cloud database connection strings managed via ``.env`` files.
- **Input Validation:** Rigorous sanitization and validation on both frontend forms and Express backend API endpoints.
- **CORS Protection:** Express CORS configuration restricting unauthorized cross-origin API calls.

9. DEPLOYMENT

Layer	Deployment Platform / Target
Frontend Hosting	Netlify / Vercel (Static Web App)
Backend API	Render / Node.js Host
Database	Neon PostgreSQL Cloud Database
Live URL	https://amfstudio.in/
Production Setup	Vite static build served alongside Node.js/Express REST API connected to Neon PostgreSQL.

10. TIMELINE & CONCLUSION

PROJECT TIMELINE

Phase	Completion Date
Requirement Analysis	08-06-2026
UI/UX Design	09-06-2026
Development Phase	10-06-2026
Testing & Deployment	02-07-2026

CONCLUSION

AMF STUDIO digital platform delivers a modern, reliable, and high-performance solution for photography and videography session scheduling and client management. The full-stack architecture ensures smooth client booking, cloud PostgreSQL database resilience, and effortless administrative control across desktop and mobile devices.

11. PROJECT COST

Component	Cost (INR)
Website Development Cost	Rs. 5,000/-
Domain Cost	Rs. 1,000/-
Total Project Cost	Rs. 6,000/-

12. SCREENSHOTS & VISUAL PREVIEWS

Figure 1: Home Hero Section Preview



Figure 2: Booking & Slot Selection UI Preview



Figure 3: Portfolio Media Gallery Preview



Figure 4: About Studio Section Preview



13. APPROVED & SIGN OFF

PREPARED BY	REVIEWED BY	APPROVED BY
Date: _____	Date: _____	Date: _____